

SkyDelay: Your Smart Flight Guide

Anticipate aviation chaos with **Machine Learning driven insights** and real-time weather data — built for passengers, analysts, and airlines alike.

Prediction Module (*Public Access*)

The "**Live Flight Predictions**" list uses our KNN Classifier model to analyze upcoming flights before they leave the gate. Enter your flight number or origin airport to filter the live list instantly.

None (On Time)

Estimated delay of **0–15 min.** You're good to go!

Low

Delay of **15–30 min.** Usually recovered during flight.

Moderate

Delay of **30–60 min.** Check your tight connections.

Severe

Over **60 min.** High risk of congestion or adverse weather.

💡 Predictions update dynamically as wind speeds and visibility conditions change at the departure airport.

What is OTP?

OTP (On-Time Performance) is the percentage of flights that depart or arrive within the scheduled time. The **1-OTP** metric represents the delay index used to color-code airports on the map.

Map Color Legend

Color	Status	Meaning
Cian	Fluid	Avg. delay under 15 min. Normal operations.
Blue	Congest ed	Avg. delay 15–30 min. Possible taxiway waits.
Purple	Alert	Avg. delay exceeds 30 min. Degraded operations.

Interactive Dashboard (*Business Subscription*)

Advanced tools for analysts, airlines, and logistics companies focused on historical performance and deeper operational insight.



Historical Delay Series

Track delay trends over time to identify recurring disruptions, seasonal effects, and long-term performance shifts.



Weather Correlations & Heatmaps

Analyze how temperature, wind, and visibility relate to delays, with heatmaps that reveal the most frequent disruption patterns.



OTP by Airport

Compare on-time performance across airports to understand where operations are running smoothly and where delays are accumulating.

Dashboard Tabs

Overall

Consolidated view of system-wide metrics and trends.

Airports

Deep dive into individual airport performance and OTP breakdowns.

Weather

Detailed weather correlations and their impact on delays.

Flights

Historical flight list with details such as delay, aircfrat model, etc.

Color Themes

